

SecLMNER: A framework for enhanced named entity recognition in multi-source cybersecurity data using large language models

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ARTICLE INFO

Keywords:

Cybersecurity
Named entity recognition
Large language models
Open-source intelligence

ABSTRACT

In the realm of cybersecurity, Named Entity Recognition (NER) has predominantly centered on cyber threat intelligence. However, cybersecurity-related information often resides in open-source intelligence and unprocessed tool outputs. With structured, semi-structured, and unstructured text data coexisting, traditional BERT-based NER methods encounter challenges in handling diverse data formats and technical terminology richness. To tackle these obstacles, this paper presents a framework that integrates large language models for NER in multi-source cybersecurity data. Leveraging decoder-based large language models' generative capabilities, the framework intelligently crafts natural language segments containing data information for enhanced adaptability. Subsequently, the SecureBERT model, excelling as the premier open-source solution for NER in the network security domain, leverages a pre-trained encoder tailored for cybersecurity text to proficiently detect diverse entities and associated information embedded within natural language segments. The experimental findings demonstrate that our proposed method outperforms traditional BERT and its variants in fine-tuning across five distinct cybersecurity text data sources. The results reveal that by integrating three generative language models, each with parameters under 10 billion, we achieve an improvement in Recall ranging from 8.1% to 21.81% over the state-of-the-art open-source SecureBERT, and an increase in F1 score from 6.19% to 16.7%. Moreover, with the enhancements to our framework, the NER performance can match that of open-source large-parameter language models.

1. Introduction

Named Entity Recognition (NER) is a crucial task within the field of Natural Language Processing (NLP). Its primary objective is to identify entities with specific meanings in a text, such as names of people, locations, organizations, and other proper nouns. NER has widespread applications in various domains, including information extraction, text classification, machine translation, and question-answering systems. This task encompasses both entity extraction and entity classification. By fine-tuning pre-trained models (Lu et al., 2023; Raffel et al., 2020) based on the Transformer architecture, the performance of NER has seen significant improvements.

Despite the notable advancements in NER technology, current models still exhibit significant limitations when applied to specialized domains. General NER models are typically trained on diverse and broad datasets, but they often perform poorly when handling domain-specific terminology and context. This issue is particularly pronounced

in fields such as cybersecurity, medicine, and law, where the vocabulary and entity types are highly specialized and complex. One of the primary challenges is the lack of comprehensive and representative annotated datasets for these specialized domains. Without sufficient domain-specific data, models cannot learn the unique patterns and entity relationships inherent in these fields. Additionally, the complexity of professional language, including jargon, abbreviations, and highly technical terms, further complicates the accurate identification and classification of entities by the models.

Taking the cybersecurity domain as an example, open-source intelligence (OSINT) sources are extensive and varied, including but not limited to social media, blogs, technical forums, news websites, threat intelligence reports, and academic papers. The quality of this information varies greatly, with both structured and unstructured data formats coexisting. These sources contain a plethora of technical terms, abbreviations, and specialized names, such as RPC, SQL, and APT,

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<https://doi.org/10.1016/j.eswa.2025.126651>

Received 21 October 2024; Received in revised form 30 December 2024; Accepted 20 January 2025

Available online 28 January 2025

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which require the model to possess corresponding domain knowledge. Additionally, cybersecurity tools like intrusion detection systems and network traffic monitoring tools output data in diverse formats, including structured data (e.g., JSON), semi-structured data, and unstructured data, posing challenges for data integration and standardization.

Current NER approaches in cybersecurity largely rely on fine-tuned BERT models (Gao et al., 2021). While these models perform well on well-preprocessed text data, their effectiveness diminishes when dealing with complex and varied data formats. Recent advancements in NLP with Decoder-Only architectures like GPT (Brown et al., 2020) and Llama models (Touvron et al., 2023), as well as Encoder-Decoder architectures like GLM (Du et al., 2022), have introduced new breakthroughs. These models excel in tasks related to question answering and dialogue generation. Researchers are exploring the potential of these large language models to leverage the knowledge acquired during training. However, these models are primarily designed for text generation tasks, whereas NER has traditionally been approached as a sequence labeling task. The inconsistency of the target tasks might lead to poor performance when using generative models to address the problem of NER.

To address these challenges, this paper proposes a framework that combines large language models to tackle NER in the cybersecurity domain with multi-source data. The framework employs a Decoder-based large language model to leverage its powerful generation capabilities and intelligently generate natural language segments containing data information, ensuring good adaptability. Simultaneously, considering the widespread hallucination issue of large language models, we have added a verification step. The large language model self-assesses the quality of the generated text segments, checks whether important information is omitted, and regenerates if necessary, thereby alleviating the hallucination phenomenon. Subsequently, fine-tuning is conducted on SecureBERT (Aghaei et al., 2022) pre-trained with texts in the cybersecurity domain, ultimately completing the identification and output of named entities. This enables the Encoder-based large language model to effectively identify various entities and relevant information within the generated natural language segments.

Experimental results demonstrate that our approach outperforms traditional methods using BERT and its variants fine-tuning across five different sources of cybersecurity text data. Our method not only identifies entities recognized by Previous open-source NER models but also uncovers more specialized entities, significantly improving detection performance.

2. Related work

In this section, we will delve into a variety of methods for named entity recognition, the evolution of named entity recognition in the cybersecurity domain, and the assistance and transformation that generative large language models have brought to the field of named entity recognition.

In recent years, Named Entity Recognition has faced significant challenges due to the diversity of entity types and the inherent complexity of the entities themselves. Different types of entities possess distinct characteristics and contextual environments, necessitating the design of specialized models and algorithms for their recognition. With the advent of deep learning technologies, deep learning-based NER models (Jehangir et al., 2023; Li et al., 2022; Yadav & Bethard, 2018) have provided new solutions for this task. These models can automatically learn the features and contextual information of entities, thereby achieving high-precision entity recognition. Currently, common deep learning NER models include those based on Convolutional Neural Networks (CNN) (Parsaieimehr et al., 2023), Recurrent Neural Networks (RNN) (Lin et al., 2021), and Transformer models. Generally, NER problems are addressed using sequence labeling methods, with various algorithms and datasets designed accordingly. For instance, the BERT (Kenton & Toutanova, 2019) model can be used to obtain

token-level representations for token classification. Additionally, deep neural networks such as bidirectional LSTM (Liu, Xu et al., 2019), Conditional Random Fields (Lafferty et al., 2001) (CRF), and models combining bidirectional LSTM and CRF (Lample et al., 2016) can capture contextual information, thereby enhancing prediction results.

Building on BERT's capability for segment understanding and sequence labeling, RoBERTa (Liu, Ott et al., 2019) employs larger model parameters, more extensive data, and longer training times, enabling it to generalize better to downstream tasks compared to BERT. LBERT (Warikoo et al., 2021), a lexicon-aware Transformer-based bidirectional encoder representation model, explores local and global contextual representations in sentence-level classification tasks, showing significant improvements over BioBERT (Lee et al., 2020) in extracting biological entity relationships. FinBERT-MRC (Zhang & Zhang, 2023) aims to extract substantial financial knowledge from unstructured texts by transforming the task into a machine reading comprehension problem.

In the cybersecurity domain, SecureBERT (Aghaei et al., 2022) is pre-trained on a large corpus of cybersecurity texts using RoBERTa, with modifications to the pre-training weights. APTNER (Wang et al., 2022) is a comprehensive named entity recognition dataset, manually annotating named entities in the cybersecurity threat intelligence domain from APT reports of various cybersecurity companies, defining 21 types of cybersecurity-related entities. LDNCTI (Huang et al., 2023) is also an NER dataset for cybersecurity threat intelligence, addressing the issue of label imbalance by introducing a Transformer model with a triplet loss and a ranking gradient coordination mechanism (TSGL). CTIBERT (Park & You, 2023) has aggregated a cybersecurity corpus from a variety of reputable data sources, predominantly comprising documents in HTML and PDF formats. By precisely extracting cyber threat intelligence, it facilitates enhanced performance of NLP tasks within the cybersecurity domain.

Generative large models have demonstrated unparalleled capabilities in understanding and utilizing natural language (Chowdhery et al., 2023; Perez et al., 2021; Vilar et al., 2023). In the context of NER, they can assist in directly identifying entities and annotating NER datasets. GPT-NER (Wang et al., 2023) employs prompt engineering and few-shot learning to transform sequence labeling tasks into generative tasks, marking entities in text with special tags. However, it can only label one segment at a time with one tag, and the tags themselves are relatively generic, lacking specialization for specific domains. NuNER (Bogdanov et al., 2024) uses data annotated by large models to create a data-efficient, compact foundational model specifically for NER, facilitating the customization of NLP models in the future.

3. Large language models

Within this section, we will introduce three distinct large language model architectures based on the Transformer framework: Encoder-only, Decoder-only, and Encoder-Decoder. Each architecture boasts unique characteristics, leading to varied capabilities and suitability for different tasks. Gaining a comprehensive understanding of the relationship between model architecture and task requirements is instrumental in constructing the SecLMNER framework.

The emergence and development of large language models have significantly advanced the field of natural language processing and have driven the growth and dissemination of artificial intelligence (AI) technologies. By leveraging pre-training and fine-tuning strategies, large language models can learn rich linguistic representations from vast amounts of textual data. Further fine-tuning on specific tasks enables these models to achieve superior performance in those tasks. In this paper, we will introduce various large language models based on three different architectures and discuss their respective characteristics.

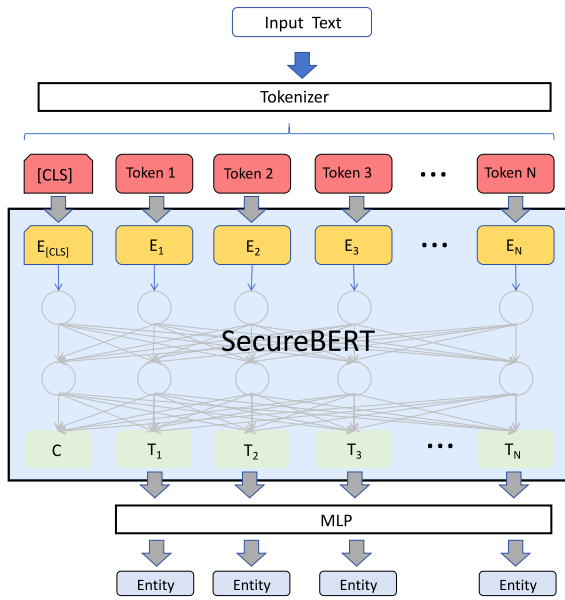


Fig. 1. SecureBERT architecture for pre-training against masked words. SecureBERT model contains 12 hidden layers and 12 attention heads, where the size of each hidden state has the dimension of 768, and the input embedding dimension is 512, the same with RoBERTa.

3.1. Encoder-only

Encoder-Only models are deep learning models that focus on understanding and analyzing input information. The primary task of these models is to encode the input data into an internal representation, rather than generating new content. Due to their focus on comprehension and analysis, Encoder-Only models excel in various understanding and classification tasks. For instance, in text classification tasks, Encoder-Only models effectively convert input text into feature vectors and classify based on these vectors. This makes them particularly useful in applications such as spam detection and news categorization. Encoder-Only models are widely applied in other natural language processing tasks such as named entity recognition, sentence similarity computation, and question-answering systems. Due to their robust comprehension capabilities, these models can extract valuable information from complex language environments, thereby supporting a wide range of applications.

Currently, Encoder-Only models predominantly leverage BERT and RoBERTa as foundational architectures for research. RoBERTa represents an enhanced version of the original BERT model. BERT relies on random masking and the prediction of tokens. The original BERT implementation performs a single masking during data preprocessing, resulting in a static mask. In contrast, RoBERTa employs dynamic masking, where a new masking pattern is generated each time a sequence is input into the model. This approach allows the model to adapt to various masking strategies and learn diverse linguistic representations as a continuous stream of data is processed. In terms of model scale, RoBERTa encompasses a larger parameter set. During training, it experiments with a range of batch sizes from 256 to 8000, and it is trained on a significantly larger dataset compared to BERT. While BERT was trained on the 16 GB BookCorpus dataset along with English Wikipedia, RoBERTa utilizes a corpus of 160 GB of raw text, including the CC-NEWS dataset. In the context of named entity recognition tasks, experimental results with CTIbert indicate that SecureBERT achieves the best performance among open-source models, with a Precision of 73.47, a Recall of 72.51, and an F1 score of 72.99. This surpasses the performance of SecBERT and SecRoBERTa, as well as that of bert-base-uncased and roberta-base. SecureBERT continues the training from

the Roberta-base model, utilizing their compiled corpus and a custom tokenizer. Fig. 1 illustrates the architecture of the SecureBERT model. Similar to the RoBERTa architecture, the SecureBERT model comprises 12 hidden layers and 12 attention heads, with each hidden state having a dimension of 768 and an input embedding dimension of 512.

3.2. Decoder-only

In contrast to Encoder-Only models, Decoder-Only models are deep learning models that focus on content generation and creation. The primary task of these models is to generate new output content based on input prompts or contextual information. Due to their generative capabilities, Decoder-Only models excel in various generation tasks.

For instance, in text generation tasks, Decoder-Only models can produce coherent and meaningful text based on a given starting point or theme. This makes them particularly useful in applications such as automated writing, dialogue generation, and content creation. The impressive conversational and generative abilities of large language models like ChatGPT demonstrate the outstanding performance and significant potential of Decoder-Only models in generative and creative tasks.

In the experiments conducted within this paper, we have selected the open-source models Llama and Mistral for our study. These models are among the publicly available Decoder-Only architectures.

3.3. Encoder-decoder

The Encoder-Decoder architecture operates through the synergistic collaboration of its constituent encoder and decoder components. The encoder distills critical information from the input data, condensing it into a context vector, which is subsequently interpreted by the decoder to generate the desired output. This integrated mechanism facilitates efficient transformation between input and output, demonstrating robust processing capabilities and adaptability across various domains such as machine translation, text summarization, and conversational systems.

3.4. Hallucination

Encoder-Only models are primarily employed for understanding and classification tasks, where hallucination issues are relatively rare. However, in certain scenarios, such as text summarization or information extraction tasks, the model may misinterpret the input content, leading to inaccurate summaries or incorrect information extraction. Decoder-Only models are more prone to hallucination issues in generative tasks because they need to generate new content based on the given context. If the contextual information is insufficient or the model is inadequately trained, the generated content may not align with the actual situation. Encoder-Decoder models are widely used in tasks such as machine translation, text summarization, and dialogue generation, but they are also susceptible to hallucination problems. This is particularly evident when handling long texts or complex grammatical structures, where the model may produce content that is inconsistent with the input.

In NER tasks, when integrating models with a Decoder architecture, it is crucial to provide sufficient information to mitigate the occurrence of hallucination phenomena.

4. Method

In this section, we will provide a detailed exposition of the proposed framework, SecLMNER, along with the various open-source cybersecurity data sources involved. We will delineate SecLMNER into three distinct modules, each of which will be elaborated upon separately. The collaborative interaction between these modules ultimately facilitates the named entity recognition from multi-source cybersecurity data.

Table 1

Occurrence of Eight Cybersecurity Entity Types Across Five Data Sources. This table presents a comprehensive analysis of the prevalence of eight distinct cybersecurity entity types within five different data sources. The data underscores the diverse informational content inherent in each source.

Entity	IP	OS	Prot	Tool	Dom	Org	Loc	Vul
Censys	✓	✓	✓	✓	✓	✓	✓	
Dig	✓	✓	✓	✓	✓			
Shodan	✓	✓	✓	✓	✓	✓	✓	✓
Nmap	✓	✓	✓	✓	✓			✓
Whois	✓				✓	✓	✓	

4.1. Data

In conducting NER in the field of cybersecurity, the integration of multi-source data is crucial. These data sources not only enrich the diversity of information but also enhance the accuracy and comprehensiveness of the recognition results. In this paper, we primarily utilize data from Open Source Intelligence and outputs from cybersecurity tools.

Firstly, OSINT is an indispensable part of cybersecurity analysis. We collected data from Whois,¹ Censys,² and Shodan.³ The Whois database provides information about domain registrations, including registrant contact information, registration dates, and domain status. This information plays a significant role in identifying network entities and tracing the origins of malicious activities. Shodan is a search engine specifically designed for searching and analyzing devices connected to the internet. Known as the “search engine for the Internet of Things”, Shodan can scan and record information about network devices globally, including servers, routers, cameras, IoT devices, and more. Shodan provides information such as device IP addresses, open ports, service versions, and operating systems. Censys is an internet search engine aimed at helping researchers and security experts understand the security status of internet devices. Developed by a research team at the University of Michigan and launched in 2015, Censys scans the internet and records device configurations and vulnerability information, providing a wealth of security intelligence data. Both Shodan and Censys offer detailed information about internet devices, which holds significant value in OSINT. Through these platforms, security researchers can acquire data on device configurations, open ports, service versions, etc., on a global scale, which is crucial for identifying and analyzing potential security threats.

Regarding cybersecurity tools, this paper incorporates outputs from dig and nmap as part of the data sources. Dig (Domain Information Groper) is a command-line tool used for querying DNS (Domain Name System) information. It is a commonly used tool among network administrators and security researchers, capable of providing detailed DNS records for specific domains. Nmap (Network Mapper) is an open-source network scanning and security auditing tool. It is widely used for discovering devices and services on a network and assessing network security. It plays an important role in penetration testing and network management. In this paper, during the process of considering entity extraction for cybersecurity data, entities that have a strong connection with network-related data are mainly considered. The entity label types are defined as eight categories: IP addresses, operating system information, protocol information, cybersecurity tools, URLs, domain names, related organizations, and vulnerabilities. Thus, the target networks or hosts of open-source intelligence or cybersecurity tools can be described relatively comprehensively. The types of entities associated with each data source are not uniform; we present the entity types associated with each data source in Table 1. We can find that each data source

contains different cybersecurity information, which also indicates that multiple perspectives and data need to be considered when engaging in cybersecurity practices.

4.2. SecLMNER

In this section, our framework SecLMNER will be introduced. The framework is divided into three modules: the generation module, the verification module, and the recognition module. The framework accomplishes the parsing and recognition of multi-source cybersecurity data by leveraging the generation capabilities of Decoder-type models and the understanding and embedding capabilities of Encoder-type models. In the generation module, large models, trained with vast amounts of data, can understand various types of texts and express multi-source cybersecurity data more effectively. The introduction of prompt engineering enables large models to generate texts that better align with the task objectives. In the verification stage, the large model conducts self-verification of the generated content, ensuring the completeness and accuracy of the generated content. In the recognition module, fine-tuned Encoder types are used for tasks such as word segmentation and encoding of the generated text, and the identified entities and labels are ultimately output through the model and classification. The overall structure of the framework is shown in Fig. 2.

In the generation module, large models are utilized for text standardization. The raw data collected from various open-source intelligence sources and cybersecurity tools are input into a large model with a Decoder architecture. Due to the excellent generation capabilities of the Decoder architecture-based large model, this model aims to handle complex text data and transform it into coherent natural language paragraphs. This step is crucial for standardizing the data and making it more amenable for subsequent analysis. In the Decoder-only architecture, also known as the autoregressive generation model, a text generation model that does not require an encoder to preprocess the input text. The prompt is directly taken as the input, starting from the initialization state, and the text is generated step by step, with each step relying on the text generated in the previous step and the model's own internal state. Our prompt is designed as the functional requirement plus the source data text. The model will generate statements related to the prompt. The Encoder–Decoder architecture encodes the input prompt into a fixed-length vector that contains the semantic information of the prompt. The vector is then input into the encoder, and the decoder generates the text step by step. At present, both the Encoder–Decoder and Decoder-Only models have strong generation capabilities and are capable of perceiving context during generation, understanding semantics and structure, and producing text that is coherent and adaptable. As of now, there is no definite conclusion on which architecture is superior. Therefore, in the subsequent experiments, we conducted experiments on models of both architectures.

In the verification module, large models are employed for checking information omissions. The text generated in the previous step may have omitted information or have low relevance to the target text. Therefore, a link of checking and reflection by the large model is added. The implementation approach is to have the large language model evaluate whether the generated content omits too much information. If there is excessive omission, the text is regenerated. When the maximum number of iterations is reached or there is no omitted information, the final text segment is output. Fig. 3 provides an example of a standard conversion from the output of NMAP to a natural language paragraph without information omission. It is observed that employing few-shot learning enhances the efficiency of the validation module. The Nmap data in the figure represents the results of a port scan on a single IP address. Each port's status and the corresponding service are displayed on a new line, which is not in the form of fluent natural language. When drafting the prompt, we need to define the model's role as a penetration testing analyst. This helps the model better adapt to and understand the text. We then need to describe the requirement to

¹ <https://www.whois.com/whois/>.

² <https://search.censys.io/>.

³ <https://www.shodan.io/>.

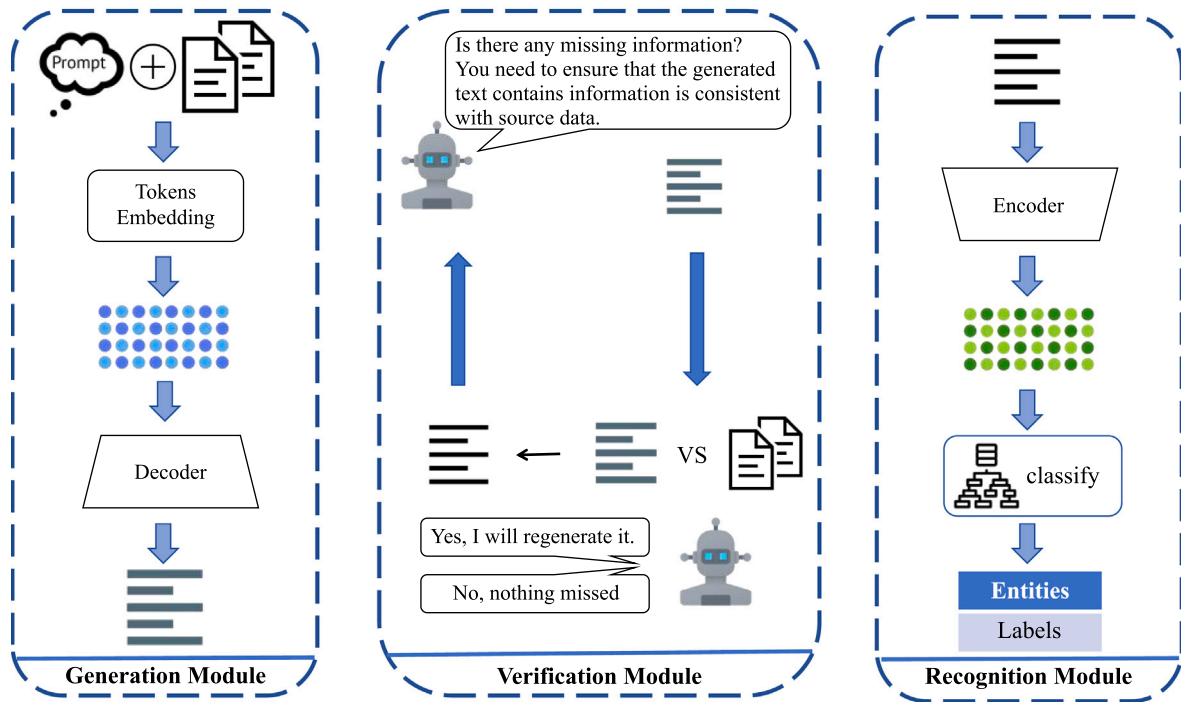


Fig. 2. Framework of SeclMNER. The generation module is responsible for transforming multi-source, heterogeneous cybersecurity text into coherent and highly readable natural language segments. The verification module is tasked with verifying and refining the text conversion processes within the generation module. The recognition module is dedicated to efficiently performing named entity recognition on segments that are semantically clear and informationally complete.

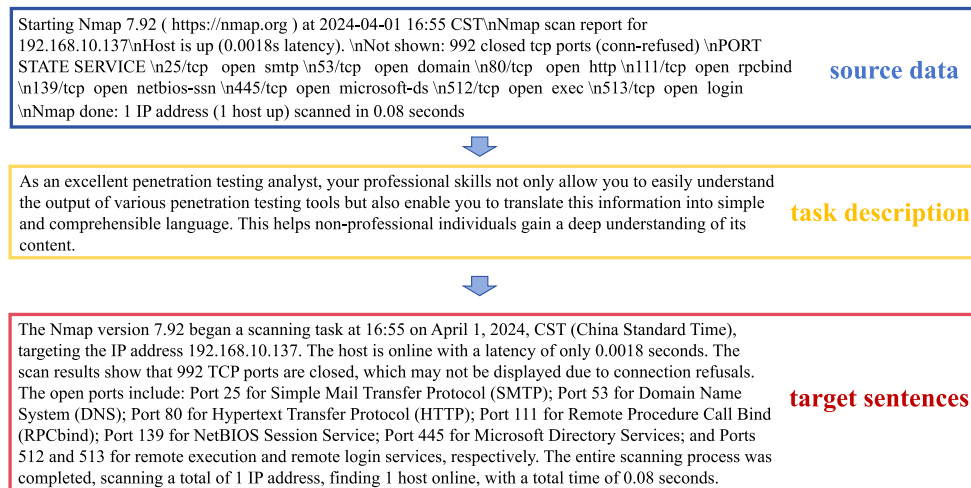


Fig. 3. Example of converting source data into a natural passage. By crafting detailed task descriptions as prompt templates and applying the same set of prompts to various cybersecurity texts, we facilitate the generation of high-quality, semantically clear texts, which in turn aids in the subsequent recognition of entities.

convert the information into natural language. The target sentence should, as shown in the figure, clearly and coherently describe the Nmap scan process in natural language, presenting information about each port and its corresponding service with both full names and abbreviations. This provides rich contextual information for the next phase of recognition.

In the recognition module, a large language model is utilized for entity recognition. Thanks to the meticulous preprocessing and transformation steps in the earlier stages, the final output is expected to have a high degree of accuracy. In our review of related work, we observed that among various open-source models, SecureBERT stands out as the most effective model for entity recognition in the cybersecurity domain.

Therefore, in the experimental section, we opt to leverage SecureBERT as the base model and fine-tune it for the subtask of named entity recognition.

Interaction between each module is conducted through natural language. Any form of open-source data is directly input into the generation module without preprocessing. The generation module outputs comprehensible and easily understandable natural language segments. The validation module then processes the natural language segments produced by the generation module, generating new natural language segments as output. Finally, these segments are input into the recognition module, which directly outputs the entity type corresponding to each subword in the natural language segments. By linking these three

Table 2

The performance of multiple open-source models fine-tuned under identical conditions on the APTNER dataset was evaluated. Among these, SecureBERT exhibited the superior performance.

Model name	Precision	Recall	F1	Accuracy
Bert-base-uncased	0.5875	6455	0.6151	0.9273
Roberta-base	0.5939	6578	0.6242	0.9350
Spanbert	0.5627	6061	0.5836	0.9295
Spanbert-large	0.5746	6203	0.5966	0.9263
SecBert	0.5474	6166	0.5799	0.9214
SecRoberta	0.5469	6187	0.5806	0.9271
SecureBERT	0.6076	6779	0.6408	0.9311

modules sequentially, we achieve a more comprehensive extraction of cybersecurity named entities from open-source data.

5. Experiment

In this section, we will discuss the experiments conducted on the SecLMNER framework. Section 5.1 covers model selection and preparation, where we identify the optimal BERT variant model. In Section 5.2, we compare generative language models with varying parameter counts on the same named entity recognition task to discern any differences. Section 5.3 showcases the capabilities of our proposed SecLMNER framework by contrasting it with the current state-of-the-art open-source cybersecurity named entity recognition model, SecureBERT. In Section 5.4, we design ablation studies to validate the contributions of each component within the SecLMNER framework. Section 5.5 is devoted to a comprehensive experimental setup that integrates language models ranging up to 10 billion parameters, as well as more advanced online language models, to showcase the efficacy of our framework. Additionally, this section delves into an examination of the framework's limitations, providing insights into the constraints and challenges encountered during the experimental process. Finally, in Section 5.6, we discuss the computational resources required by the framework.

5.1. Model and data preparation

In the generation module and verification module, in order to demonstrate that our method is applicable to multiple models and fully integrates the capabilities of large language models. We conducted tests on the open-source models GLM-4, LLaMA3-7b, and Mixtral-7 × 8b. GLM-4 is a large-scale language model developed by Zhipu AI. It utilizes the GLM model structure proposed by Tsinghua University's KEG Laboratory, positioning itself as a robust tool in natural language processing tasks. LLaMA3, created by Meta AI, is an open-source large language model. It has shown exceptional performance in several industry-standard benchmark tests, particularly in conversational applications. Pre-trained with over 150 trillion tokens from selectively sourced public online data, LLaMA 3 ensures comprehensive coverage and high-quality output. Mixtral-7 × 8b, developed by Mistral AI, is an advanced artificial intelligence model based on the Transformer architecture. It incorporates a Mixture-of-Expert layer, a special layer that plays a pivotal role in processing input data. Each data point is assigned to two out of eight processing units, referred to as "experts". This design maximizes computational resources, significantly enhancing the model's efficiency and performance when handling vast datasets.

Among these models, we utilized the aforementioned models via the Hugging Face online chat interface⁴ as well as the official ChatGLM website.⁵ Additionally, we elected to employ the smaller-parameter

variants of these models, namely GLM-4-9B, Mistral-8B, and LLaMA3-8B. For the purpose of inference, these open-source models were selected for local deployment. GLM follows an Encoder-Decoder architecture, whereas Llama and Mixtral utilize a Decoder-only architecture. We employed the approach of utilizing APIs to invoke these models and conduct experiments.

In our study, we selected the APTNER dataset (Wang et al., 2022), a relatively recent and entity-type-appropriate cybersecurity named entity recognition dataset, which encompasses a variety of entity types. Utilizing the same training data, our objective was to identify the most capable variant of the BERT model. The models under consideration included BERT-base-uncased, Roberta-base, SpanBERT (Joshi et al., 2020), SpanBERT-Large, SecureBERT (an enhanced version of RoBERTa), SecBERT,⁶ and SecRoBERT, both of which are modifications based on BERT and RoBERTa, respectively. These models were sourced from the Hugging Face platform. We compared the performance of these models on a test set after fine-tuning with the selected open-source dataset. During the training process, we set the learning rate to 2e-05, the batch size to 8, the seed to 42, and employed the AdamW optimizer for 5 epochs of training.

In experiments involving NER, the effectiveness of a model is typically assessed using three key metrics: Precision, Recall, and the F1 Score. Precision is the ratio of correctly identified named entities to the total number of entities that the model predicted. It measures the model's ability to avoid false positives. In other words, it reflects the proportion of positive identifications that were actually correct. Mathematically, Precision is defined as:

$$Precision = \frac{TP}{TP + FP} \quad (1)$$

Recall is the ratio of correctly identified named entities to the total number of actual entities in the text. It measures the model's ability to avoid false negatives, indicating how many of the actual entities were correctly identified. Mathematically, Recall is defined as:

$$Recall = \frac{TP}{TP + FN} \quad (2)$$

The F1 Score is the harmonic mean of Precision and Recall. It provides a single metric that balances the trade-off between Precision and Recall. An F1 Score closer to 1 indicates a better performance of the model, with both high Precision and high Recall, while an F1 Score closer to 0 indicates poorer performance. The F1 Score is calculated as follows:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (3)$$

The results are depicted in the accompanying Table 2. It is evident that SecureBERT emerges as the optimal model, achieving the highest scores across all metrics. This superior performance can be attributed to the fact that SecureBERT was trained on a corpus of cybersecurity documents selected during the pre-training phase, utilizing the RoBERTa base architecture. This specialized training enhanced the model's ability to process text in the cybersecurity domain.

Consequently, we have selected SecureBERT as the Encoder-only model within the SecLMNER framework. The tagging scheme was configured to be of the BIO (Begin-Inside-Outside) type. SecureBERT's tokenizer decomposes words into smaller subwords for identification, which enhances the model's performance and generalization capabilities. However, the number of entities identified is not the actual number of entities but rather the number of subentities. Further processing is required to yield the final entity recognition results.

In terms of experimental data preparation, we collected data from the results of the shodan, censys, whois databases, and the dig command for 20 different IPs or domain names, as well as the results of 20 nmap commands with different commands and targets, as the source data. The detailed information regarding data is provided in the

⁴ <https://huggingface.co/chat>.

⁵ <https://chatglm.cn>.

⁶ <https://huggingface.co/jackaduma/SecBERT>.

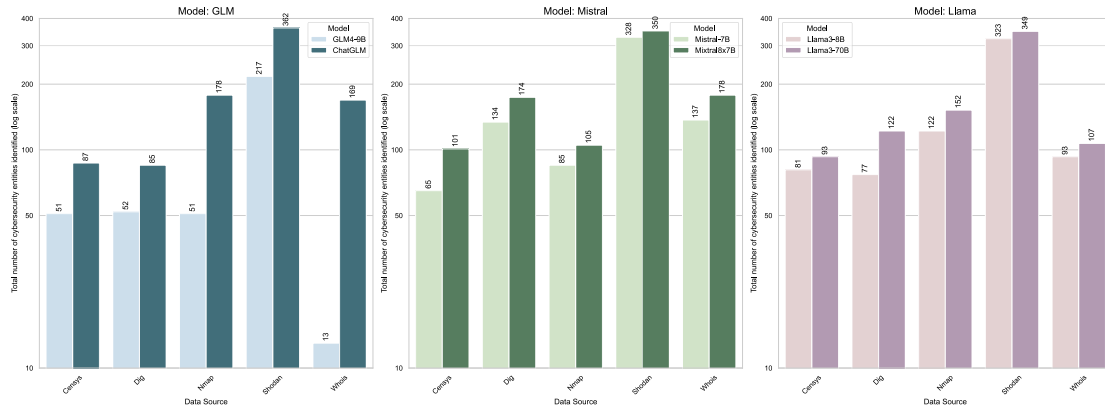


Fig. 4. Comparative performance analysis of LLMs with varying parameter sizes in cybersecurity NER. This figure illustrates a performance comparison of the GLM, Llama, and Mistral models across different parameter sizes in the context of cybersecurity named entity recognition. The graph clearly demonstrates that models with larger parameter counts consistently outperform their smaller-parameter counterparts.

“Data Availability” section. To verify the effectiveness of the proposed framework, we followed the traditional NER method and directly used the fine-tuned SecureBERT to tokenize the source data and embed vectors for recognition. This was used for a comparative experiment with the framework for NER to compare the identified entities.

Given that different generative models can produce sentences with variations in word order, vocabulary, and grammar under the same conditions, it is unnecessary to annotate named entities for every segment generated by these models. To evaluate the improvement in NER performance when incorporating large language models, we will manually consolidate the NER results processed by the large language models. We will then identify and count the correctly recognized entities that are pertinent to the cybersecurity domain. Finally, we will compare the number of valid entities identified across various models in different types of cybersecurity texts.

5.2. Model size impact

In the performance of large language models on text processing tasks, a general pattern that emerges is that an increase in parameter size correlates with enhanced model capabilities (Patil & Gudivada, 2024). Numerous complex challenges that were insurmountable for many models have been addressed by scaling up the parameter count, thereby enhancing the potential for problem resolution. In the context of cybersecurity named entity recognition, we conducted an experimental study. We uniformly prompted the models with the instruction: “I require you to perform cybersecurity named entity recognition, encompassing eight categories: IP, OS, Protocol, Tool, Domain, Organization, Location, and Vulnerability. Please generate the output in this sequence and refrain from producing any superfluous information”. By tabulating the total number of named entities identified from 20 data samples across five distinct data sources, we observed in Fig. 4 that for the same model architecture, a higher parameter count generally exhibits superior performance in comprehending intentions and extracting cybersecurity entities. Additionally, it can be observed that the performance gap between language models of two different parameter scales is not consistent across various data sources. This discrepancy underscores the distinct nature of the models and their varying performance on different data sources, suggesting that the recognition effectiveness may be influenced by the specific entity types and the data sources themselves. The specific details will be discussed in Section 5.5.

5.3. SecLMNER vs. Traditional NER

To provide a more granular exposition of the Named Entity Recognition process, we segmented the data from Censys, Dig and meticulously

Table 3

Recognition results on Censys data. In terms of the total number of entities recognized, SecLMNER exhibits a significant improvement over the Encoder-Only models when conducting NER on the Censys data source.

Method	Recognized/Total number			
	1–5	6–10	11–15	16–20
Finetuned SecureBERT	11/48	9/39	4/37	16/45
SecLMNER-GLM4-9B	36/48	33/39	25/37	33/45
SecLMNER-Llama3-8B	33/48	26/39	23/37	33/45
SecLMNER-Mixtral-7B	33/48	30/39	25/37	37/45

Table 4

Recognition results on Dig data. SecLMNER also shows enhancements over traditional NER methods in the Dig data, albeit less prominently than in other datasets.

Method	Recognized/Total number			
	1–5	6–10	11–15	16–20
Finetuned SecureBERT	25/38	20/32	16/29	16/30
SecLMNER-GLM4-9B	24/38	24/32	19/29	25/30
SecLMNER-Llama3-8B	26/38	21/32	19/29	18/30
SecLMNER-Mixtral-7B	33/38	27/32	24/29	23/30

identified the total number of entities that should be recognized, as well as the total number of entities correctly identified by each method within each segment. Subsequently, we conducted a comparative analysis of the different approaches. Tables 3 and 4 illustrate the relationship between the number of entities identified and the total number of entities present. It was observed that, in every segment of data, the framework we proposed consistently outperformed in processing the Censys data or Dig data, a trend that was mirrored across other datasets.

In the experiments using the Censys data source, the number of enhanced entities ranged from 75 to 87, with an improvement in detection rate from 44% to 51%. In the experiments using the Dig data source, the number of enhanced entities ranged from 7 to 28, with an improvement in detection rate from 5% to 20%. The significant improvement in NER for the Censys data can likely be attributed to the fact that Censys data is in JSON format, which lacks sufficient contextual information for encoder-based large models to understand. In contrast, the reason the framework shows less improvement in NER on Dig data compared to Censys data is that Dig data is already closer to natural language paragraphs. Additionally, the text generated by decoder-based large models provides contextual information that can enhance recognition effectiveness.

Table 5 presents the recognition outcomes for each method across five distinct data sources. The experimental results indicate that our NER approach, integrated with Large Language Models, exhibits commendable performance with open-source cybersecurity data, significantly outperforming BERT-based models in terms of the number of

Table 5

Recognized number on multisource data. SecLMNER demonstrates varying degrees of improvement over traditional cybersecurity NER methods in identifying total entities across five distinct data sources, highlighting the efficacy of our proposed framework.

Method	Censys	Delta	Shodan	Delta	Whois	Delta	Dig	Delta	Nmap	Delta
Finetuned SecureBERT	40	0	159	0	64	0	77	0	73	0
SecLMNER-GLM4-9B	127	87↑	341	182↑	184	120↑	92	15↑	150	77↑
SecLMNER-Llama3-8B	115	75↑	331	172↑	182	118↑	84	7↑	174	101↑
SecLMNER-Mixtral-7B	125	85↑	361	202↑	235	171↑	105	28↑	212	139↑

Table 6

The performance of SecLMNER and fine-tuned SecureBERT across five distinct data sources. It can be observed that SecLMNER consistently outperforms the fine-tuned SecureBERT in terms of efficacy on each of the data sources evaluated.

Methods	Precision	Recall	F1
Finetuned-SecureBERT	0.9419	0.5341	0.6817
SecLMNER-GLM4-9B	0.9401	0.6151	0.7436
SecLMNER-LLaMa3-8B	0.9673	0.7068	0.8168
SecLMNER-Mixtral-8B	0.9736	0.7522	0.8487

entities identified. The model adeptly converts formatted and unformatted data into natural language segments and leverages its contextual comprehension to extract the required cybersecurity entities.

We integrated the data to obtain the performance of the four models across all datasets. The data is presented in Table 6. Upon observation, SecLMNER has improved F1 scores compared to the current state-of-the-art open-source NER model, SecureBERT, with an increase of more than 6.19%, which demonstrates the effectiveness of the framework. This means that when generating natural language text based on given input, generative large models tend to produce content that aligns with the provided text. Coupled with an information verification step against the original text, this reduces the occurrence of hallucinations. Furthermore, this approach helps to elucidate the implicit information within open-source data, thereby enhancing information extraction and subsequent utilization.

In this experimental study, both the Encoder–Decoder and Decoder-Only model architectures have demonstrated their formidable generative capabilities in the domain of Named Entity Recognition. They adeptly managed data of various structures, yielding natural language segments that were articulate and with clearly defined focal points. Upon synthesizing the outcomes from the processing of diverse datasets, we did not discern any significant differences in the generative efficacy between the Encoder–Decoder and Decoder-Only architectures. However, the former necessitates greater resources during training and data processing, a consideration that must be taken into account in environments with constrained resources. This insight is crucial for balancing the trade-offs between performance and resource allocation in the deployment of NER models.

5.4. Ablation study

To investigate the impact of various components within our proposed framework on overall performance, we conducted an ablation study. The rationale behind the ablation study design was to systematically remove or replace specific modules within the model, thereby observing the concrete effects of these alterations on model performance. We conducted experiments on the SecureBERT model fine-tuned solely for the task, on three language models incorporating only the Decoder architecture, and on the same models' NER framework for the identification of eight cybersecurity entity types. The total number of entities identified was compared across these setups.

As shown in Fig. 5, We found that when the language models with the Decoder architecture were reduced in their capability to process multi-source data into a coherent natural language segment, there was a decrease of more than 7% in the total number of named entities identified, highlighting the necessity of preprocessing heterogeneous

data before identification. When the fine-tuned SecureBERT was minimized to rely solely on the language model for comprehension and extraction, there was at least a 2% reduction in the total number of named entities, indicating that RoBERTa is efficient and cost-effective in terms of computational resources.

The results of the ablation study indicate that each component within our proposed framework exerts a varying degree of influence on the ultimate performance. SecureBERT remains superior to 7B or 8B language models in terms of reduced computational resource consumption, while these language models, in turn, enhance the model's performance in cybersecurity entity recognition tasks by processing multi-source heterogeneous data. These insights provide valuable guidance for further refinement of our framework.

5.5. SecLMNER's competitiveness and limitations

To accentuate the practicality and computational resource conservation of our proposed framework, we conducted a boxplot analysis comparing models with parameters below 10B to larger parameter models accessed online and those enhanced by our framework in Figs. 6–8. The analysis reveals that, even with a marked reduction in computational resources, the models augmented by our framework exhibit superior and more stable performance, approximating the capabilities of powerful large language models. This phenomenon is consistently observed across the three categories of models tested in our experiments.

To comprehensively evaluate the performance of our proposed framework and uncover potential shortcomings, we aggregated the recognition results by entity category. We computed the performance metrics for four classifications across eight distinct entity types. The experimental outcomes are presented in Tables 7 and 8, revealing varying extraction capabilities among different entity types across multiple models. We observed that the SecLMNER framework generally enhances recognition effectiveness for most entities, with Domain and Vulnerability being notable exceptions. Specifically, the F1 score increases by over 18.84% for IP addresses, by more than 28.7% for operating systems, by over 29.93% for protocols, by more than 26.49% for tools, by 48.25% for organizations, and by over 32.69% for locations. In our detailed analysis, we found that when the source data contains multiple similar entity categories listed together, some domain names and vulnerability titles may still be omitted despite the use of prompt engineering to guide the generative model against missing information. This occurs because within the selected data sources, the Whois database frequently lists Name Server domain names, such as those for Instagram.com, which appear as follows: Name Server: C.NS.INSTAGRAM.COM; Name Server: B.NS.INSTAGRAM.COM; Name Server: D.NS.INSTAGRAM.COM; Name Server: A.NS.INSTAGRAM.COM. These listings exhibit a significantly higher entity density compared to other text segments, with the entities being quite similar to one another. When generating models process and count these segments, there is a tendency to use “etc.” to represent the entire list. A similar pattern is observed in Shodan data with the enumeration of potential CVE numbers associated with hosts. Subsequently, the recognition module is unable to supplement this information, contradicting our original intention of utilizing generative models to complement and elaborate on information. We are considering employing more sophisticated prompt engineering in future iterations for improvement, such as determining the type of data source and specifying the format of the text to be extracted.

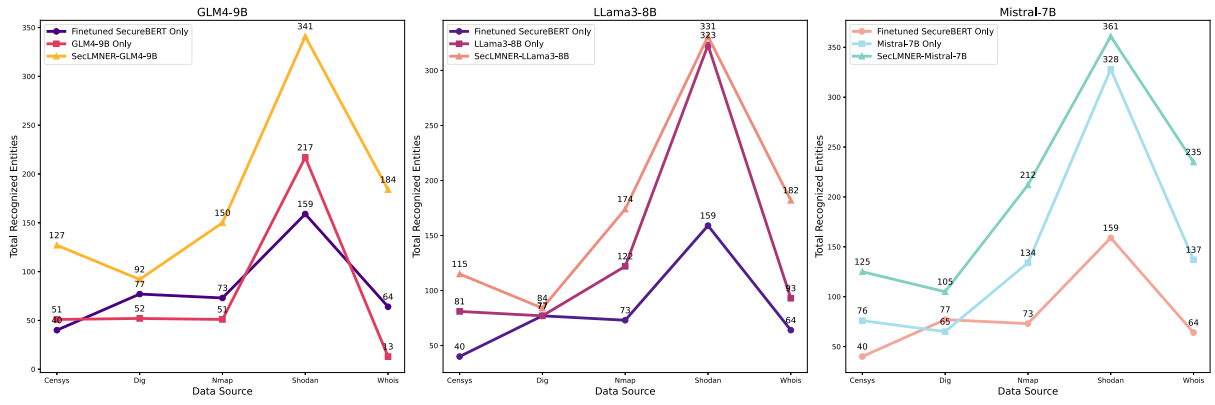


Fig. 5. Ablation study. The figure illustrates the results obtained from conducting ablation experiments using individual large language models and SecureBERT for NER. It is observed that the performance of both models exhibits varying degrees of decline compared to SecLMNER. This finding suggests that each component within our framework contributes to the overall performance of the framework.

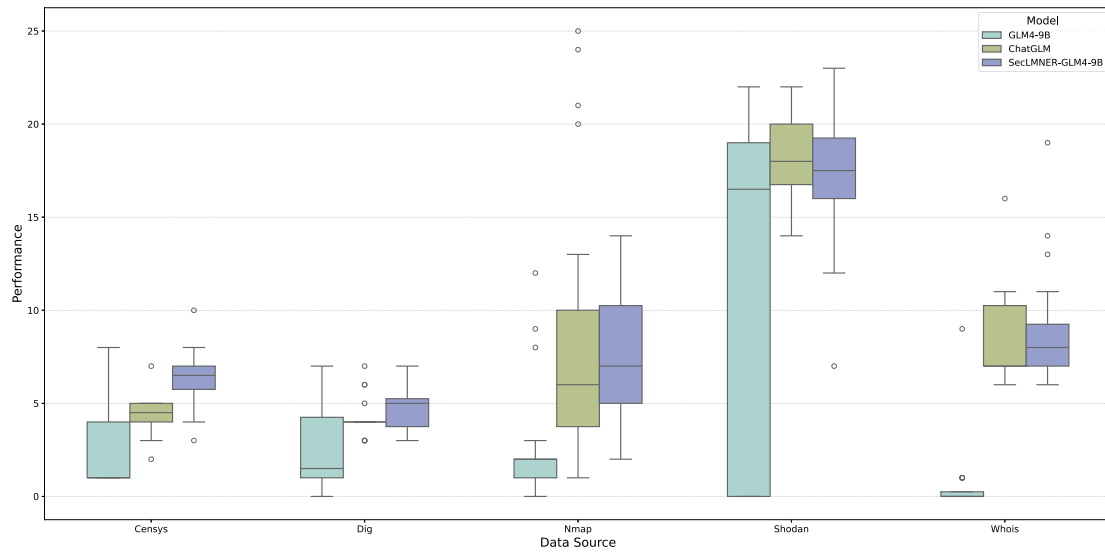


Fig. 6. Enhanced performance of SecLMNER with GLM4-9B. The figure demonstrates that through effective organization of the framework, GLM4-9B exhibits greater stability and superior performance on Cybersecurity NER tasks compared to ChatGLM, even when solely leveraging RoBERTa-based model as the encoder.

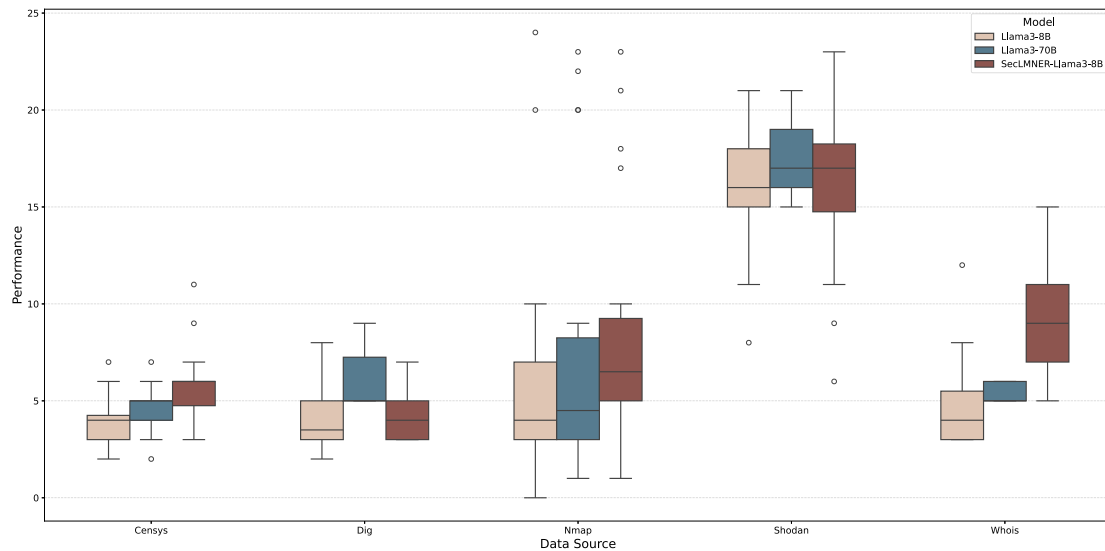


Fig. 7. Enhanced performance of SecLMNER with Llama3-8B. The figure illustrates that by incorporating SecLMNER with Llama3-8B, the performance surpasses that of Llama3-8B alone in NER tasks. Across various data sources such as Censys, Nmap, and Whois, the average and overall performance of SecLMNER with Llama3-8B outperforms even Llama3-70B.

Table 7

The recognition performance of 4 methods on 4 different types of entities (IP addresses, Operating Systems, Protocols, and Tools). P represents Precision, and R represents Recall score. Considering the F1 score comprehensively, it can be observed that the framework's performance on multi-source cybersecurity data shows an improvement over the best-performing open-source model, SecureBERT.

Methods	IP			OS			Protocol			Tool		
	P	R	F1	P	R	F1	P	R	F1	P	R	F1
Finetuned-SecureBERT	0.9677	0.4196	0.5854	0.9091	0.2273	0.3636	0.9167	0.33	0.4853	0.8730	0.44	0.5851
SecLMNER-GLM4-9B	0.9744	0.6496	0.7795	0.9762	0.7736	0.8631	0.9391	0.7660	0.8437	0.8615	0.8358	0.8485
SecLMNER-LLaMa3-8B	0.9810	0.7803	0.8692	0.9	0.5094	0.6506	0.9722	0.6731	0.7955	0.9302	0.8	0.8602
SecLMNER-Mixtral-8B	0.9459	0.75	0.8367	0.9474	0.5455	0.6923	0.9829	0.7099	0.8244	0.9497	0.7692	0.8500

Table 8

The recognition performance of 4 methods on other different types of entities (Domain, Vulnerability, Organization, and Location).

Methods	Domain			Vulnerability			Organization			Location		
	P	R	F1	P	R	F1	P	R	F1	P	R	F1
Finetuned-SecureBERT	0.8961	0.6053	0.7225	0.9748	0.5446	0.6988	0.9333	0.2692	0.4179	0.8333	0.1887	0.3077
SecLMNER-GLM4-9B	0.9697	0.4286	0.5944	0.9897	0.5581	0.7138	0.8636	0.7917	0.8261	0.9714	0.5231	0.6800
SecLMNER-LLaMa3-8B	0.9907	0.4798	0.6465	0.9854	0.9439	0.9642	0.9000	0.7200	0.8000	0.9667	0.4833	0.6444
SecLMNER-Mixtral-8B	0.9867	0.5944	0.7419	0.9862	0.9954	0.9098	0.9286	0.7500	0.8298	0.9737	0.6981	0.8132

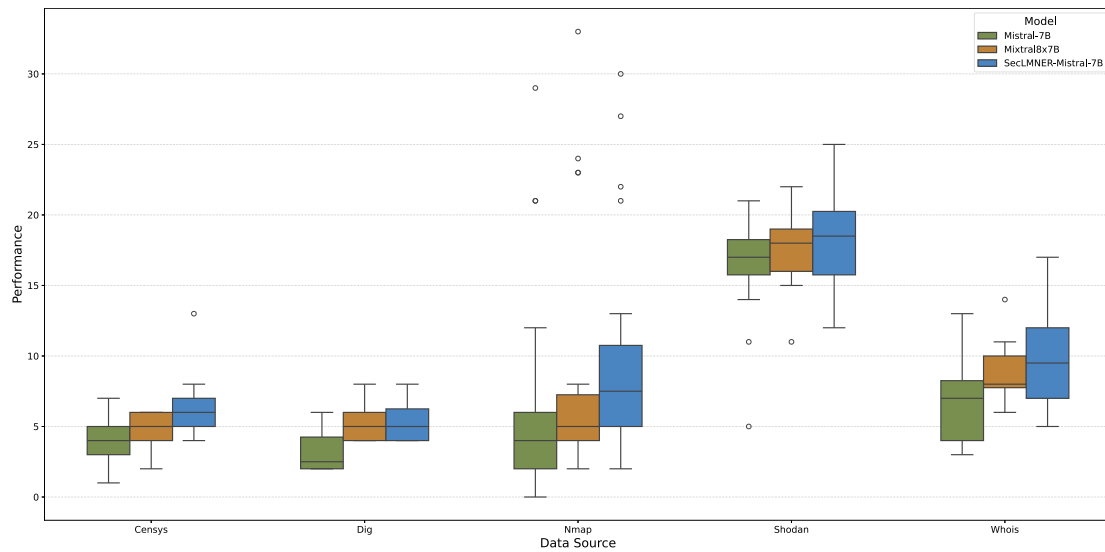


Fig. 8. Enhanced performance of SecLMNER with Mistral-7B. The figure demonstrates that the inclusion of SecLMNER with Mixtral-7B enables performance to surpass that of Mixtral-7B alone in NER tasks. Across five different data sources, SecLMNER with Mixtral-7B outperforms Mixtral-7B in both average and overall performance.

5.6. Computational expense discussion

During the fine-tuning phase, we trained BERT and its variants using NVIDIA GeForce RTX 4070 32 GB. Training times ranged from 5 min to 43 min. However, when utilizing generative large language models for ensuring good interaction and coherence, larger computational resources are necessary. For inference, we employed an A800 80 GB graphics card. We documented the inference times of generative large language models within the 10B parameter range during the experimental process. The results are as follows: Llama3-8B averaged 5.79 s per inference, Mixtral averaged 4.92 s, and GLM4-9B averaged 6.81 s per inference. The differences were minimal. For the remaining larger generative models, we utilized online API calls.

Furthermore, the aforementioned experiments did not demonstrate superior performance of the Encoder-Decoder structured GLM model over the other two models in the task of cybersecurity entity recognition. However, through our proposed framework, which integrates fine-tuned SecureBERT with models containing up to 10 billion parameters, recognition performance can match or surpass that of larger models while significantly reducing computational expenditure. This is because the resources required for fine-tuning models like Llama and GLM are substantially greater than those needed for fine-tuning models such as BERT and RoBERTa. By combining the fine-tuning of RoBERTa-like models with the direct application of language models within the

10 billion parameter range, the framework not only achieves a marked improvement in cybersecurity entity recognition performance but also mitigates computational resource consumption.

6. Conclusion and future work

This paper proposes a framework for network security information extraction by integrating large language models. By understanding the similarities and differences among various architectures of large language models, a multi-model framework is designed that leverages the distinct characteristics and strengths of each model to enhance task performance. This approach effectively utilizes the advantages of different models to assist in the recognition and processing of multi-source network security text information. Experimental results demonstrate that the proposed framework improves the performance of traditional named entity recognition, which often relies on single data structures or sources. By enabling the understanding and transformation of multi-source data through large models, the framework enhances both adaptability and accuracy in entity recognition. It achieves an improvement in Recall ranging from 8.1% to 21.81% over the state-of-the-art open-source SecureBERT, and an increase in F1 score from 6.19% to 16.7%. Moreover, experimental discussions demonstrate that through the reinforcement provided by this framework, the performance of

Table 9
Instructions for Obtaining 20 Records from an Nmap Data Source.

Usage	Command
Simple host discovery	nmap IP_address/IP_segment
Detect the OS of a host:	nmap -O IP_address
Scan a host using TCP connect	nmap -sT IP_address
Scan a host using TCP SYN scan	nmap -sS IP_address
Scan a list of hosts from a file	nmap -iL file_name
Scan a range of hosts	nmap IP_range
Scan a specific port	nmap -p port IP_address
Scan multiple specific ports	nmap -p ports IP_address
Scan 100 most common ports	nmap -F IP_address
Scan all 65535 TCP ports	nmap -p - IP_address
Scan using TCP ACK:	nmap -sA IP_address
Scan using UDP scan	nmap -sU IP_address
Perform an aggressive scan	nmap -A IP_address
Output the results to a file:	nmap IP_address -oN file_name
Combine scan techniques:	nmap -sS -sU -p ports IP_address
Scan an IPv6 address:	nmap -6 IPv6_address
Scan for vulnerabilities:	nmap --script=vuln IP_address
Guess and print out hostnames:	nmap -sY IP_address
Scan for specific vulnerabilities	nmap -sV -p port --script=vulname IP
Scan and detect services versions:	nmap -sV IP_address

language models within the 10B parameter range can rival that of high-performance large language models such as Llama3-70B, thereby reducing the resource consumption associated with directly employing large language models for NER tasks.

Future work can be improved in the following areas: collecting a broader range of network security text data from diverse types and sources to cover a wider array of entity types, thereby increasing the framework's specialization. Further efforts can also be directed towards information extraction tasks such as relation extraction, constructing a comprehensive network security knowledge graph based on multi-source open intelligence. There is still room for improvement in named entity extraction effectiveness, suggesting a need for deeper exploration of model architectures. Simplifying the framework and incorporating more considerations specific to the cybersecurity domain can further improve the efficiency and accuracy of network security information extraction (see Table 9).

CRedit authorship contribution statement

Yunlong Zhang: Writing - Original draft preparation, Writing - Review & Editing, Software. **Jingju Liu:** Methodology, Supervision. **Xiaofeng Zhong:** Data curation, Conceptualization. **Lei Wu:** Investigation, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request. Regarding the Nmap data, usage, and the corresponding 20 commands are provided in the Table 9. The execution environment for these commands is within a set of virtual machines configured by the author for research purposes only. The Whois and DIG data are the search results from the respective databases for 20 well-known domain names: baidu.com, google.com, microsoft.com, cnn.com, bing.com, imdb.com, FACEBOOK.com, youtube.com, amazon.com, twitter.com, wikipedia.org, instagram.com, linkedin.com, netflix.com, reddit.com, pinterest.com, WHATSAPP.com, tumblr.com, apple.com, ebay.com.

References

- Aghaei, E., Niu, X., Shadid, W. G., & Al-Shaer, E. (2022). SecureBERT: A domain-specific language model for cybersecurity. In F. Li, K. Liang, Z. Lin, & S. K. Katsikas (Eds.), *Lecture notes of the institute for computer sciences, social informatics and telecommunications engineering: Vol. 462, Security and privacy in communication networks - 18th EAI international conference, SecureComm 2022, virtual event, October 2022, proceedings* (pp. 39–56). Springer.
- Bogdanov, S., Constantin, A., Bernard, T., Crabbé, B., & Bernard, E. (2024). NuNER: Entity recognition encoder pre-training via LLM-annotated data. In Y. Al-Onaizan, M. Bansal, & Y. Chen (Eds.), *Proceedings of the 2024 conference on empirical methods in natural language processing, EMNLP 2024, Miami, FL, USA, November 12–16, 2024* (pp. 11829–11841). Association for Computational Linguistics, Retrieved from <https://aclanthology.org/2024.emnlp-main.660>.
- Brown, T. B., Mann, B., Ryder, N., Subbiah, M., Kaplan, J., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A., Agarwal, S., Herbert-Voss, A., Krueger, G., Henighan, T., Child, R., Ramesh, A., Ziegler, D. M., Wu, J., Winter, C., ..., Amodei, D. (2020). Language models are few-shot learners. In H. Larochelle, M. Ranzato, R. Hadsell, M. Balcan, & H. Lin (Eds.), *Advances in neural information processing systems 33: annual conference on neural information processing systems 2020, NeurIPS 2020, December 6–12, 2020, virtual*. Retrieved from <https://proceedings.neurips.cc/paper/2020/hash/1457c0d6bfc4967418bfb8ac142f64a-Abstract.html>.
- Chowdhery, A., Narang, S., Devlin, J., Bosma, M., Mishra, G., Roberts, A., Barham, P., Chung, H. W., Sutton, C., Gehrmann, S., Schuh, P., Shi, K., Tsvyashchenko, S., Maynez, J., Rao, A., Barnes, P., Tay, Y., Shazeer, N., Prabhakaran, V., ..., Fiedel, N. (2023). PaLM: Scaling language modeling with pathways. *Journal of Machine Learning Research*, 24, 240:1–240:113.
- Du, Z., Qian, Y., Liu, X., Ding, M., Qiu, J., Yang, Z., & Tang, J. (2022). GLM: general language model pretraining with autoregressive blank infilling. In S. Muresan, P. Nakov, & A. Villavicencio (Eds.), *Proceedings of the 60th annual meeting of the association for computational linguistics (volume 1: long papers), ACL 2022, Dublin, Ireland, May 22–27, 2022* (pp. 320–335). Association for Computational Linguistics, <http://dx.doi.org/10.18653/v1/2022.ACL-LONG.26>, Retrieved from <https://doi.org/10.18653/v1/2022.acl-long.26>.
- Gao, C., Zhang, X., Han, M., & Liu, H. (2021). A review on cyber security named entity recognition. *Frontiers of Information Technology & Electronic Engineering*, 22(9), 1153–1168.
- Huang, Y., Su, M., Xu, Y., & Liu, T. (2023). NER in cyber threat intelligence domain using transformer with TSL. *Journal of Circuits, Systems and Computers*, 32(12), 2350201:1–2350201:16.
- Jehangir, B., Radhakrishnan, S., & Agarwal, R. (2023). A survey on named entity recognition — datasets, tools, and methodologies. *Natural Language Processing Journal*, 3, Article 100017. <http://dx.doi.org/10.1016/j.nlp.2023.100017>, Retrieved from <https://www.sciencedirect.com/science/article/pii/S2949719123000146>.
- Joshi, M., Chen, D., Liu, Y., Weld, D. S., Zettlemoyer, L., & Levy, O. (2020). SpanBERT: Improving pre-training by representing and predicting spans. *Transactions of the Association for Computational Linguistics*, 8, 64–77. http://dx.doi.org/10.1162/TACL_A_00300, Retrieved from https://doi.org/10.1162/tacl.a_00300.
- Kenton, J. D. M.-W. C., & Toutanova, L. K. (2019). Bert: Pre-training of deep bidirectional transformers for language understanding. vol. 1, In *Proceedings of Naacl-HLT (p. 2)*. Minneapolis, Minnesota.
- Lafferty, J. D., McCallum, A., & Pereira, F. C. N. (2001). Conditional random fields: Probabilistic models for segmenting and labeling sequence data. In C. E. Brodley, & A. P. Danyluk (Eds.), *Proceedings of the eighteenth international conference on machine learning (ICML 2001), Williams College, Williamstown, MA, USA, June 28 - July 1, 2001* (pp. 282–289). Morgan Kaufmann.
- Lample, G., Ballesteros, M., Subramanian, S., Kawakami, K., & Dyer, C. (2016). Neural architectures for named entity recognition. In K. Knight, A. Nenkova, & O. Rambow (Eds.), *NAAACL HLT 2016, the 2016 conference of the North American chapter of the association for computational linguistics: human language technologies, San Diego California, USA, June 12–17, 2016* (pp. 260–270). The Association for Computational Linguistics.
- Lee, J., Yoon, W., Kim, S., Kim, D., Kim, S., So, C. H., & Kang, J. (2020). BioBERT: a pre-trained biomedical language representation model for biomedical text mining. *Bioinformatics*, 36(4), 1234–1240.
- Li, J., Sun, A., Han, J., & Li, C. (2022). A survey on deep learning for named entity recognition. *IEEE Transactions on Knowledge and Data Engineering*, 34(1), 50–70. <http://dx.doi.org/10.1109/TKDE.2020.2981314>.
- Lin, J. C., Shao, Y., Djenouri, Y., & Yun, U. (2021). ASRNN: a recurrent neural network with an attention model for sequence labeling. *Knowledge-Based Systems*, 212, Article 106548.
- Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., Levy, O., Lewis, M., Zettlemoyer, L., & Stoyanov, V. (2019). RoBERTa: A robustly optimized BERT pretraining approach. CoRR abs/1907.11692. arXiv:1907.11692. Retrieved from <http://arxiv.org/abs/1907.11692>.
- Liu, W., Xu, T., Xu, Q., Song, J., & Zu, Y. (2019). An encoding strategy based word-character LSTM for Chinese NER. In J. Burstein, C. Doran, & T. Solorio (Eds.), *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, NAAACL-HLT 2019, Minneapolis, MN, USA, June 2–7, 2019, volume 1 (long and short papers)* (pp. 2379–2389). Association for Computational Linguistics.

- Lu, J., Zhu, D., Han, W., Zhao, R., Namee, B. M., & Tan, F. (2023). What makes pre-trained language models better zero-shot learners? In A. Rogers, J. L. Boyd-Graber, & N. Okazaki (Eds.), *Proceedings of the 61st annual meeting of the association for computational linguistics (volume 1: long papers)*, ACL 2023, Toronto, Canada, July 9-14, 2023 (pp. 2288–2303). Association for Computational Linguistics, <http://dx.doi.org/10.18653/v1/2023.ACL-LONG.128>, Retrieved from <https://doi.org/10.18653/v1/2023.acl-long.128>.
- Park, Y., & You, W. (2023). A pretrained language model for cyber threat intelligence. In M. Wang, & I. Zitouni (Eds.), *Proceedings of the 2023 conference on empirical methods in natural language processing: EMNLP 2023 - industry track, Singapore, December 6-10, 2023* (pp. 113–122). Association for Computational Linguistics.
- Parsaeimehr, E., Fartash, M., & Torkestani, J. A. (2023). Improving feature extraction using a hybrid of CNN and LSTM for entity identification. *Neural Processing Letters*, 55(5), 5979–5994.
- Patil, R., & Gudivada, V. (2024). A review of current trends, techniques, and challenges in large language models (LLMs). *Applied Sciences*, 14(5).
- Perez, E., Kiela, D., & Cho, K. (2021). True few-shot learning with language models. In M. Ranzato, A. Beygelzimer, Y. N. Dauphin, P. Liang, & J. W. Vaughan (Eds.), *Advances in neural information processing systems 34: annual conference on neural information processing systems 2021, NeurIPS 2021, December 6-14, 2021, virtual* (pp. 11054–11070).
- Raffel, C., Shazeer, N., Roberts, A., Lee, K., Narang, S., Matena, M., Zhou, Y., Li, W., & Liu, P. J. (2020). Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*, 21(140), 1–67.
- Touvron, H., Lavril, T., Izacard, G., Martinet, X., Lachaux, M., Lacroix, T., Rozière, B., Goyal, N., Hambro, E., Azhar, F., Rodriguez, A., Joulin, A., Grave, E., & Lample, G. (2023). LLaMA: Open and efficient foundation language models. <http://dx.doi.org/10.48550/ARXIV.2302.13971>, CoRR abs/2302.13971. arXiv:2302.13971. Retrieved from <https://doi.org/10.48550/arXiv.2302.13971>.
- Vilar, D., Freitag, M., Cherry, C., Luo, J., Ratnakar, V., & Foster, G. F. (2023). Prompting PaLM for translation: Assessing strategies and performance. In A. Rogers, J. L. Boyd-Graber, & N. Okazaki (Eds.), *Proceedings of the 61st annual meeting of the association for computational linguistics (volume 1: long papers)*, ACL 2023, Toronto, Canada, July 9-14, 2023 (pp. 15406–15427). Association for Computational Linguistics.
- Wang, X., He, S., Xiong, Z., Wei, X., Jiang, Z., Chen, S., & Jiang, J. (2022). APTNER: A specific dataset for NER missions in cyber threat intelligence field. In *25th IEEE international conference on computer supported cooperative work in design, CSCWD 2022, Hangzhou, China, May 4-6, 2022* (pp. 1233–1238). IEEE.
- Wang, S., Sun, X., Li, X., Ouyang, R., Wu, F., Zhang, T., Li, J., & Wang, G. (2023). GPT-NER: named entity recognition via large language models. CoRR abs/2304.10428. arXiv:2304.10428. Retrieved from <https://doi.org/10.48550/arXiv.2304.10428>.
- Warikoo, N., Chang, Y., & Hsu, W. (2021). LBERT: Lexically-aware transformers based bidirectional encoder representation model for learning universal bio-entity relations. *Bioinformatics*, 36(22–23), 5565.
- Yadav, V., & Bethard, S. (2018). A survey on recent advances in named entity recognition from deep learning models. In E. M. Bender, L. Derczynski, & P. Isabelle (Eds.), *Proceedings of the 27th international conference on computational linguistics* (pp. 2145–2158). Santa Fe, New Mexico, USA: Association for Computational Linguistics, Retrieved from <https://aclanthology.org/C18-1182>.
- Zhang, Y., & Zhang, H. (2023). FinBERT-MRC: financial named entity recognition using BERT under the machine reading comprehension paradigm. *Neural Processing Letters*, 55(6), 7393–7413.